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Elastic moduli of borosilicate glasses doped with heavy metal oxides

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ABSTRACT

Comparative studies on the theoretical and experimental values of elastic moduli of the borosilicate glasses (70-x) SiO_2–$30B_2O_3$–xHMOs glass system, where HMOs (heavy metal oxides) are TiO_2, BaO and Bi_2O_3 with 0, 1, 2, 3, 4 and 5 mol% of each HMO, were investigated. Elastic moduli were assessed by measuring the ultrasonic velocities. The number of network bonds per unit volume, the average of a stretching force constant, the average of cross-link density, the average of ring diameter and the theoretical bond compression bulk modulus were calculated by using a theoretical bond compression model to confirm the obtained results from the experiments. The results show that changes in the structure of the glass depend on a type and concentration of HMOs. Moreover, the experimental elastic moduli are in good agreement with the theoretical values.

1. Introduction

Glass is a solid material of interest because it is transparent to visible light and is a good insulator. Of specific interest for a glass material, borosilicate glasses have high chemical and mechanical resistance, very low electrical conductivity and thermal expansion coefficient. Thus, they are widely used in laboratory, optical, heat-resistant, fiber, pharmaceutical and sealing glass applications, and even for nuclear waste immobilization [1–4].

Elastic properties are very informative regarding the structures of glasses and they are directly related to the interatomic potentials. The glasses are isotropic and have only two independent elastic constants: longitudinal and shear moduli. These parameters are obtained from calculating the longitudinal and shear velocities and the densities of the glasses. The bulk modulus, Young's modulus and Poisson's ratio can also be deduced. The elastic moduli of borosilicate glasses containing transition, rare earth and/or heavy metal oxide (HMO) depend on ultrasonic waves at room temperature and were previously reported by several groups [4–15].

The glasses based on heavy metal oxides (HMOs) such as BaO, Bi_2O_3, PbO, TiO_2, Ag_2O, etc. have always been an area of interest because of their characteristic structural and physical properties such as high refractive index, high thermal expansion, high density, low transformation temperature and excellent infrared transmission (IR). Therefore, these glasses have been desirable aspirants for potential applications in IR technologies, design of laser devices and non-linear optics [16,17]. Among HMO glasses, bismuth and/or barium borosilicate glasses are the subject of growing and intense research. The glasses containing BaO and/or Bi_2O_3 have attracted considerable attention because of their vast range of applications in the fields of radiation shielding, glass-ceramics, reflecting windows, thermal and mechanical sensors, etc. [18–20]. In addition, the glasses containing significant concentrations of transition metal oxides (TMOs) such as TiO_2, ZnO, Fe_2O_3, V_2O_5, MnO_2, etc. are of continuing interest because of their applicability in memory switching, electrical threshold, optical switching devices, etc. [21–25].

In this work, the elastic moduli of borosilicate glasses will be discussed. Information regarding the number of network bonds per unit volume, the average of stretching force constant, the average of ring size diameter and the average of the cross-link density will be examined and discussed. The theoretical values of bond compression bulk modulus and elastic moduli will be calculated and compared with the experimental values.

2. Materials and methods

2.1. Preparation of glass samples

Rectangular shaped glass samples of the (70-x)SiO_2–$30B_2O_3$–xHMOs glass system (where HMOs are TiO_2, BaO and Bi_2O_3 with 0, 1, 2, 3, 4 and 5 mol% of each HMO) were prepared by the conventional melting technique. The oxides of SiO_2, B_2O_3, TiO_2, BaO and Bi_2O_3 used in this work were of an analytical regent grade. To prepare the glass samples, appropriate amounts of SiO_2, B_2O_3, TiO_2, BaO and Bi_2O_3 were weighed using an electronic balance with the accuracy of the order of 0.0001 g. The homogeneous mixtures were placed in ceramic crucible and melted in an electric furnace until homogeneity of the glass melt was ensured. The melted glasses were poured into graphite molds and annealed for 2 h before naturally cooling down to room temperature.

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2.2. Density and molar volume measurements

The densities of the glass samples were determined by Archimedes' principle, using n-hexane as an immersion liquid and applying the relationship (1) [6]

$$\rho = \rho_L\left(\frac{W_a}{W_a - W_b}\right) \tag{1}$$

where ρ_L is density of the immersion liquid, and where W_a and W_b are the sample weights in air and in the immersion fluid, respectively. The experiment was repeated three times to obtain an accurate value for the density. The estimated error in these measurements was about ± 0.015 g cm^{-3} (shown in Table 1). The molar volume (V_a) was calculated from the expression $V_a = \frac{M}{\rho}$, where M is the molecular weight of the glass, which was calculated according to the relationship $M = \sum x_i M_i$ [26], where x_i is the mole fraction of the component oxide i and M_i is its molecular weight. The glass packing density can be calculated from the following Eqs. (2)–(3) [7]

$$V_t = \frac{\rho}{M}\sum_i x_i V_i \tag{2}$$

where V_i is given by,

$$V_i = \frac{4\pi N_A}{3}\left(x r_M^3 + y r_O^3\right) \tag{3}$$

where N_A is Avogadro's number, and where r_M and r_O are the ionic radii of the cation and anion of the oxide M_xO_y, respectively. The uncertainties in molar volume and packing density were acquired from experiments repeated three times of densities. The estimated error in these results was about ± 0.200 cm^3·mol^{-1} and $\pm 0.003 \times 10^{-6}$ m^3, respectively (shown in Table 1).

2.3. Ultrasonic measurements and determination of elastic moduli

To measure the ultrasonic velocity in the glass samples, an ultrasonic flaw detector, SONATEST Sitescan 230, was used. The ultrasonic wave was generated from a ceramic transducer with a resonant frequency at 4 MHz and acting as a transmitter–receiver at the same time. The ultrasonic wave velocity (v) can be calculated using the following Eq. (4) [5]:

$$v = \frac{2x}{\Delta t} \quad \left(cm \cdot s^{-1}\right) \tag{4}$$

where x is the sample thickness (cm) and Δt is the time interval (s). The measurements were repeated three times to check the reproducibility of the data. The estimated error in the velocity measurements was ± 23 m·s^{-1} for the longitudinal velocity and ± 11 m·s^{-1} for the shear velocity. The elastic strain produced by a small stress can be described by two independent elastic constants, C_{11} and C_{44} [27]. Elastic moduli were calculated using the following standard relations (5)–(10) [27]:

$$\text{Longitudinal modulus } C_{11} = L = \rho v_L^2, \tag{5}$$

$$\text{Shear modulus } C_{44} = G = \rho v_S^2, \tag{6}$$

$$\text{Bulk modulus } K = L - \frac{4}{3}G, \tag{7}$$

$$\text{Young's modulus } E = (1+\sigma)2G, \tag{8}$$

$$\text{Poisson's ratio } \sigma = \frac{L-2G}{2(L-G)}, \tag{9}$$

$$\text{Microhardness } H = \frac{(1-2\sigma)E}{6(1+\sigma)}, \tag{10}$$

Debye temperature calculated from Eq. (11) [28]

$$\theta_D = \left(\frac{h}{k_B}\right)\left(\frac{3zN_A}{4\pi V_a}\right)^{1/3} v_m, \tag{11}$$

where v_L and v_S are longitudinal and transverse velocities, respectively. h is Planck's constant, k_B is Boltzmann's constant, N_A is Avogadro's number, z is the number of atoms in the chemical formula and v_m is the mean ultrasonic velocity defined by the relationship (12) [29].

$$v_m = \left[\frac{3v_L^3 v_S^3}{v_L^3 + v_S^3}\right]^{1/3} \tag{12}$$

Softening temperature T_s is related to the ultrasonic velocity of shear waves v_S by Eq. (13) [4]

$$T_s = \frac{v_S M}{C^2 Z} \tag{13}$$

where Z is the number of atoms in the chemical formula and C is the constant of proportionality and equals 507.4 $m \cdot s^{-1} \cdot K^{1/2}$. The

Table 1
Glass composition, density (ρ), molar volume (V_a) and packing density (V_t) of the glass samples.

Sample no.	SiO_2 (mol%)	B_2O_3 (mol%)	TiO_2 (mol%)	BaO (mol%)	Bi_2O_3 (mol%)	ρ (g·cm^{-3})	V_a (cm^3·mol^{-1})	$V_t \times 10^{-6}$ (m^3)
S0	70	30	0	0	0	2.548	24.704	0.485
S1–TiO_2	70	30	1	0	0	2.547	24.287	0.498
S2–TiO_2	70	30	2	0	0	2.572	23.821	0.512
S3–TiO_2	70	30	3	0	0	2.594	23.390	0.526
S4–TiO_2	70	30	4	0	0	2.614	22.985	0.540
S5–TiO_2	70	30	5	0	0	2.634	22.585	0.555
S1–BaO	70	30	0	1	0	2.582	24.933	0.485
S2–BaO	70	30	0	2	0	2.600	25.023	0.488
S3–BaO	70	30	0	3	0	2.630	25.189	0.489
S4–BaO	70	30	0	4	0	2.647	25.381	0.489
S5–BaO	70	30	0	5	0	2.671	25.503	0.491
S1–Bi_2O_3	70	30	0	0	1	2.676	24.132	0.501
S2–Bi_2O_3	70	30	0	0	2	2.800	23.645	0.516
S3–Bi_2O_3	70	30	0	0	3	2.947	23.020	0.535
S4–Bi_2O_3	70	30	0	0	4	3.021	22.996	0.540
S5–Bi_2O_3	70	30	0	0	5	3.123	22.767	0.551
The uncertainty						±0.015	±0.200	±0.003

Table 2
Longitudinal (v_L), shear (v_S) and mean (v_m) velocities, longitudinal (L), shear (G), bulk (K) and Young's (E) modulus, Poisson's ratio (σ), microhardness (H), Debye (θ_D) and softening (T_s) temperature of the glass samples.

Sample no.	v_L (m/s)	v_S (m/s)	v_m (m/s)	L (GPa)	G (GPa)	K (GPa)	E (GPa)	σ	H (GPa)	θ_D (K)	T_s (K)
S0	6134	3788	4178	95.87	36.56	47.12	87.15	0.1918	7.51	361	699
S1–TiO_2	5925	3619	3996	89.41	33.36	44.94	80.22	0.2025	6.62	347	632
S2–TiO_2	5783	3557	3925	86.02	32.54	42.63	77.82	0.1957	6.60	343	599
S3–TiO_2	5521	3393	3744	79.07	29.86	39.25	71.47	0.1965	6.04	330	535
S4–TiO_2	5399	3290	3634	76.20	28.29	38.47	68.17	0.2047	5.57	322	494
S5–TiO_2	5288	3217	3554	73.65	27.26	37.31	65.76	0.2062	5.34	316	464
S1–BaO	6057	3737	4122	94.73	36.06	46.65	86.01	0.1927	7.39	355	681
S2–BaO	5963	3655	4035	92.45	34.73	46.14	83.30	0.1991	6.97	347	656
S3–BaO	5879	3594	3968	90.90	33.97	45.60	81.64	0.2016	6.76	341	636
S4–BaO	5811	3537	3907	89.38	33.11	45.23	79.86	0.2057	6.50	335	621
S5–BaO	5734	3524	3889	87.82	33.17	43.59	79.38	0.1965	6.71	333	619
S1–Bi_2O_3	6034	3725	4109	97.43	37.13	47.92	88.53	0.1921	7.62	358	685
S2–Bi_2O_3	5938	3633	4011	98.73	36.96	49.45	88.76	0.2009	7.37	352	660
S3–Bi_2O_3	5865	3578	3952	101.37	37.73	51.07	90.82	0.2036	7.45	350	643
S4–Bi_2O_3	5791	3522	3891	101.31	37.47	51.35	90.42	0.2065	7.33	344	641
S5–Bi_2O_3	5720	3459	3824	102.18	37.37	52.36	90.56	0.2117	7.18	340	628
The uncertainty	±23	±11	±16	±0.9	±0.4	±0.6	±0.7	±0.0012	±0.08	±2	±4

uncertainties in elastic moduli, Poisson' ratio, micro-hardness, Debye temperature and softening temperature were acquired from experiments repeated three times of the densities and the ultrasonic velocities. The estimated errors in these results are shown in Table 2.

2.4. Theoretical bond compression model

A bond compression model is a useful guide for structures containing only one type of bond. For a three dimensional polycomponent oxide glass, the bond compression bulk modulus is given by Eq. (14) [30]

$$K_{bc} = \frac{n_b \overline{F}}{9} r^2 \tag{14}$$

where r is the bond length between cation and anion and n_b is the number of network bonds per unit volume of the glass given by Eq. (15) [30]

$$n_b = \frac{N_A}{V_a} \sum_i \left(x n_f \right)_i \tag{15}$$

where x is the mole fraction of the component oxide i. $\overline{F}$ is the average of stretching force constant and can be calculated from Eq. (16) [31]

$$\overline{F} = \frac{\sum \left(x n_f F \right)_i}{\sum \left(x n_f \right)_i} \tag{16}$$

where n_f is the coordination number of the cation and F is the stretching force constant of the oxide. The average atomic ring size (l) of a structure consisting of a three-dimensional network according to the ring deformation model is expressed in the form of Eq. (17) [32].

$$l = \left[0.0106 \frac{\overline{F}}{K_{exp}} \right]^{0.26} \tag{17}$$

The theoretical Poisson's ratio for the polycomponent oxide glasses according to the bond compression model is given by Eq. (18) [30]

$$\sigma_{cal} = 0.28 (\overline{n}_c)^{-0.25} \tag{18}$$

where $\overline{n}_c$ is the average cross-link density of the glass network and is given by Eq. (19) [30]

$$\overline{n}_c = \frac{1}{\eta} \sum_i (n_c)_i (N_c)_i \tag{19}$$

where n_c is the number of cross-links per cation (number of bridging bonds per cation minus two) in oxide i. N_c is the number of cations per glass formula unit and $\eta = \Sigma(N_c)_i$ is the total number of cations per glass formula unit. The theoretical bulk modulus (K_{cal}) can be calculated from Eq. (20) [31].

$$K_{cal} = 1.062 \times 10^{-29} \overline{F} l^{-4.0022} \tag{20}$$

The other theoretical elastic moduli can be obtained from the bulk modulus and Poisson's ratio for each glass system as Eqs. (21)–(23) [30].

$$G_{cal} = 1.5 K_{cal} \left[\frac{1 - 2\sigma_{cal}}{1 + \sigma_{cal}} \right] \tag{21}$$

$$L_{cal} = K_{cal} + 1.33 G_{cal} \tag{22}$$

$$E_{cal} = 2 (1 + \sigma_{cal}) G_{cal} \tag{23}$$

3. Results and discussion

3.1. Density and molar volume

The glass composition, density, molar volume and packing density are given in Table 1. The density and molar volume increase when the content of doping (TiO_2, BaO and Bi_2O_3) increases as shown in Fig. 1. The density increases with the content of doping in all the glasses. Moreover, the density of the glass samples doped with Bi_2O_3 was higher than that of the glass samples doped with BaO and TiO_2. This is due to the adding of TiO_2, BaO and Bi_2O_3 (molecular weights of 79.866, 153.326 and 465.959 $g \cdot mol^{-1}$, respectively) into the glass matrix [7].

The results of molar volume as shown in Fig. 1 revealed that the molar volumes of the glass samples decreased with the increase of content of Bi_2O_3 and TiO_2, whereas the molar volume of the glass samples doped with BaO slightly increases. The molar volume is defined as the volume occupied by the unit mass. To explain the results of molar volume the variations of packing density or compactness of the glasses

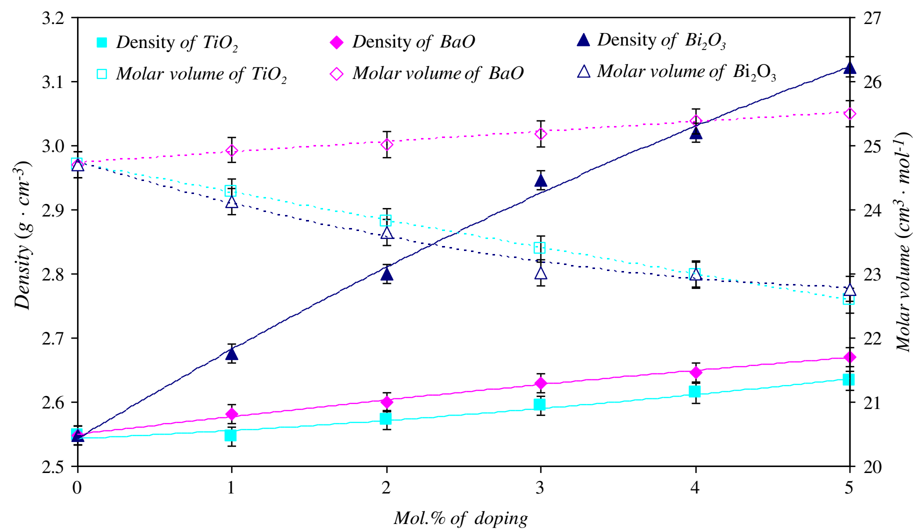

Fig. 1. Variation of densities (ρ) and molar volume (Va) of glass samples with the difference of doping (lines are drawn as guides to the eyes).

were calculated and shown in Fig. 2. The glass packing density is the ratio between the minimum theoretical volume occupied by the ions and the corresponding effective volume of the glass. The packing density or compactness of the glasses doped with Bi_2O_3 and TiO_2 shows large increase with increasing mol% of the dopant as shown in Fig. 2. These results indicate that the compactness of the glass matrix increases with the increase of the content of Bi_2O_3 and TiO_2. Therefore, the molar volume decreases with the mol% of Bi_2O_3 and TiO_2 increase. However, the compactness of the glass doped with BaO slightly increases with the increase of the content of BaO. These results suggested that the glass compactness depends on the ionic radius of the modifier. The ionic radius of the Ba^{2+} (1.49 Å) is larger than the ionic radii of Bi^{3+} and Ti^{4+} (1.17 and 0.745 Å, respectively) which can lead to an increase in the size of the interstices and increase in the molar volume [33]. Therefore, the glass doped with BaO shows the smaller increase of packing density than the glass doped with Bi_2O_3 and TiO_2. Moreover, the larger ionic radius results in increasing of the molar volume.

3.2. Ultrasonic velocity and elastic moduli

The plots of longitudinal (v_L) and shear (v_S) wave velocities in the borosilicate glasses with the mol% of the dopants are shown in Fig. 3 and exact values are shown in Table 2. It is observed that the ultrasonic wave velocities, v_L and v_S, in the glasses decrease as the mol% of the dopants (TiO_2, BaO and Bi_2O_3) increase. Preferably, the glasses doped with TiO_2 showed significant decreases of both v_L and v_S velocities with the increase of concentration of TiO_2. In general, the decrease of ultrasonic velocity is related to the increase in the number of non-bridging oxygens (NBOs) resulting in decrease in connectivity of the glass network [28]. For these reasons, adding the modifier (TiO_2, BaO and Bi_2O_3) leads to structural changes by creating the formation of NBOs. The lower sound velocity with doped TiO_2 indicates that adding TiO_2 leads to higher formation of NBOs than that of BaO and Bi_2O_3. Therefore, the decrease in ultrasonic wave velocities is due to the fact that HMO ions are involved in the glass network as modifiers by breaking up the tetrahedral bond of SiO_4 units [6]. In the asseveration of these results, the number of bonds per unit volume (n_b) is calculated by using a theoretical bond compression model and exact values are given in Table 3. It can be seen that all the samples show the decrease in the number of bonds per unit volume with the increase of the mol% of the dopants. In addition, the lower the number of bonds per unit volume of the glass doped with TiO_2, the greater the formation of non-bridging oxygens (NBOs). These results supported our discussions of the ultrasonic wave velocities of glass samples.

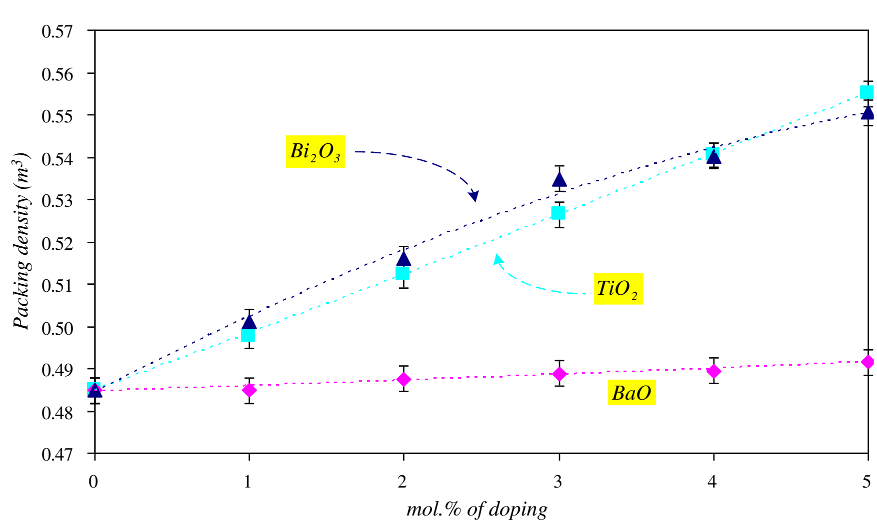

Fig. 2. Variation of packing density (Vt) of glass samples with the difference of doping (lines are drawn as guides to the eyes).

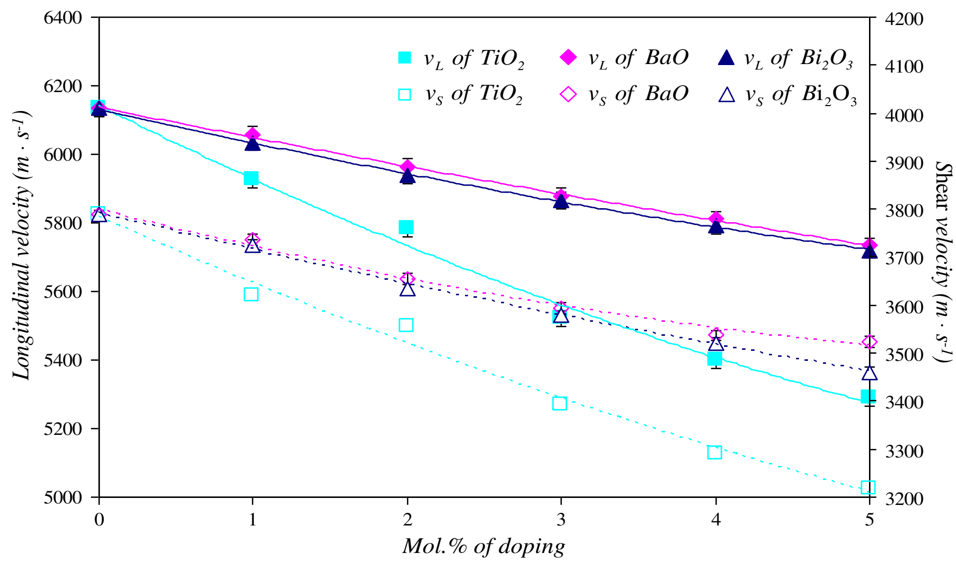

Fig. 3. Variation of longitudinal (vL) and shear (vS) velocity of glass samples with the difference of doping. The error bar is contained within the symbol size (lines are drawn as guides to the eyes).

As seen from Figs. 4–6 and Table 2, all elastic moduli values (L, G, K and E) decrease with the increase of TiO_2 and BaO concentrations while they increase with the increase of Bi_2O_3 concentration. All of the elastic moduli are related to the average strength of the bond. The average strength of the bond depends on the value of the cation–anion forces. For a given A–O–A bond angle, the A–A separation would be directly proportional to the stretching force constant $(\overline{F})$ of the glass network [34]. As the A–O–A bond force constants decrease, the energy required to produce a given degree of bond angle or length distortion and/or bond distortion decreases which leads to the decrease in the average strength of the bond. To verify these results, the stretching force constants $(\overline{F})$ are calculated by using a theoretical bond compression model, and the exact values are shown in Table 3. From the table, the average stretching force constants $(\overline{F})$ for the glass doped with TiO_2 and BaO decrease with the increase of the mol% of the dopant (TiO_2 and BaO). On the other hand, the average stretching force constants $(\overline{F})$ for the glass doped with Bi_2O_3 show the increase with the increase of the mol% of the dopant (Bi_2O_3). These results indicate that the glass doped with TiO_2 and BaO leads to the decrease in the average strength of the bond (elastic moduli were decreased) while the average strength of the bond increases when doped with Bi_2O_3 (elastic moduli were increased).

Table 3
E/G ratio, average cross-link density ($\overline{n}_c$), average stretching force constant $(\overline{F})$, average ring diameter (l), theoretical bond compression bulk modulus (K_{bc}), number of network bonds per unit volume (n_b) and K_{bc}/K_{exp} ratio of the glass samples.

Sample no.	E/G	$\overline{n}_c$	$\overline{F}$ (N/m)	l (nm)	K_{bc} (GPa)	$n_b \times 10^{22}$ (cm^{-3})	K_{bc}/K_{exp}
S0	2.384	2.615	431.5	0.5453	152.02	14.68	3.226
S1–TiO_2	2.421	2.667	429.2	0.5513	176.22	12.37	3.483
S2–TiO_2	2.421	2.657	426.8	0.5581	171.22	12.36	3.788
S3–TiO_2	2.483	2.647	424.5	0.5694	166.34	12.33	4.238
S4–TiO_2	2.516	2.636	422.3	0.5716	161.45	12.29	4.451
S5–TiO_2	2.530	2.626	420.1	0.5754	156.50	12.27	4.723
S1–BaO	2.385	2.595	430.7	0.5465	151.21	14.39	3.242
S2–BaO	2.398	2.576	429.8	0.5478	151.25	13.99	3.278
S3–BaO	2.403	2.556	429.0	0.5491	150.84	13.59	3.308
S4–BaO	2.411	2.537	428.2	0.5500	150.27	13.19	3.322
S5–BaO	2.393	2.519	427.3	0.5550	150.12	12.79	3.444
S1–Bi_2O_3	2.411	2.583	425.0	0.5449	156.52	13.88	3.266
S2–Bi_2O_3	2.440	2.552	426.3	0.5421	160.64	13.67	3.248
S3–Bi_2O_3	2.444	2.522	427.6	0.5372	165.93	13.57	3.249
S4–Bi_2O_3	2.458	2.493	428.9	0.5323	167.03	13.14	3.253
S5–Bi_2O_3	2.462	2.464	430.2	0.5312	169.64	12.80	3.240

Fig. 7 shows the variation of Poisson's ratio and micro-hardness of the glass samples as a function of the dopants and the exact values are listed in Table 2. Poisson's ratio of all samples increases as the mol% of the dopants increases. The variation of Poisson's ratio related to cross-link density decreases as the cross-link density increases. Therefore, the increase of Poisson's ratio shows that the cross-link density decreases due to the increase in the number of non-bridging oxygens after adding TiO_2, BaO and Bi_2O_3, respectively. To confirm the results, the average number of cross-links density ($\overline{n}_c$) was calculated by using the theoretical bond compression model and shown in Table 3. The average number of cross-links density ($\overline{n}_c$) decreased when the mol% of the dopants increases. These results strongly support the increase of Poisson's ratio with the increase of the mol% of the dopants because of the occurrence of non-bridging oxygens (NBOs). Moreover, these results support our explanation of the ultrasonic velocities. The micro-hardness (H) of the glass samples as a content of the dopants is shown in Fig. 7. It is defined as the resistance of a material to permanent indentation or penetration [35]. It can be seen that the micro-hardness has the same characteristic as the ultrasonic velocities with the increase of the mol% of the dopants. Therefore, it is evident that the decrease in micro-hardness is related to the decrease in the rigidity of the glass network due to the occurrence of non-bridging oxygens (NBOs) [36].

Debye temperature (θ_D) and softening temperature (T_s) of the glasses are listed in Table 2 and the plots of Debye temperature (θ_D) and softening temperature (T_s) are shown in Fig. 8. Debye temperature (θ_D) is an important parameter of a solid, describes the properties arising from atomic vibration and is directly proportional to the mean ultrasonic velocity (v_m). The variations of the mean ultrasonic velocity are shown in Table 2 [29]. Softening temperature (T_s) is another important parameter defined as the temperature point at which viscous flow changes to plastic flow [6,28]. It can be observed from Fig. 8 that both the Debye and softening temperature decrease as the mol% of the dopants increases. The decreases in the mean of ultrasonic velocity, Debye temperature and softening temperature are mainly contributed by the increase in formation of non-bridging oxygens (NBOs) as a direct effect of the insertion of HMO ions (ions of TiO_2, BaO and Bi_2O_3) into the glass network structures [6,28].

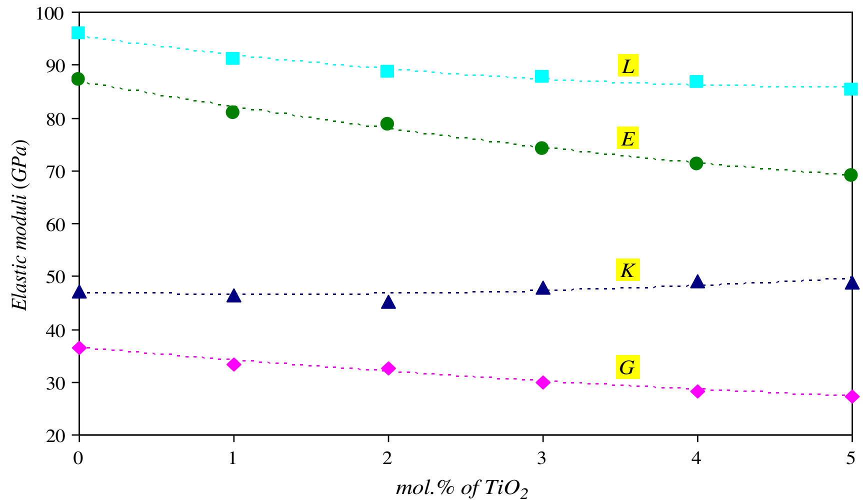

Fig. 4. Variation of elastic moduli of glass samples with different mol% of TiO2. The error bar is contained within the symbol size (lines are drawn as guides to the eyes).

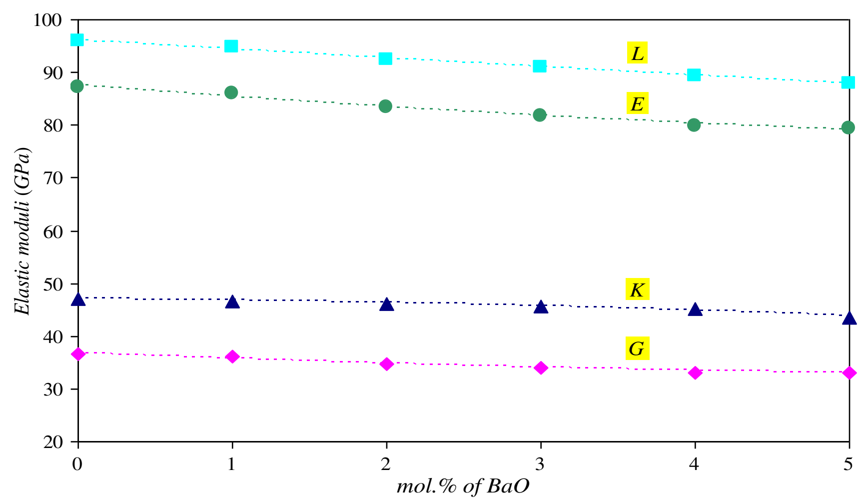

Fig. 5. Variation of elastic moduli of glass samples with different mol% of BaO. The error bar is contained within the symbol size (lines are drawn as guides to the eyes).

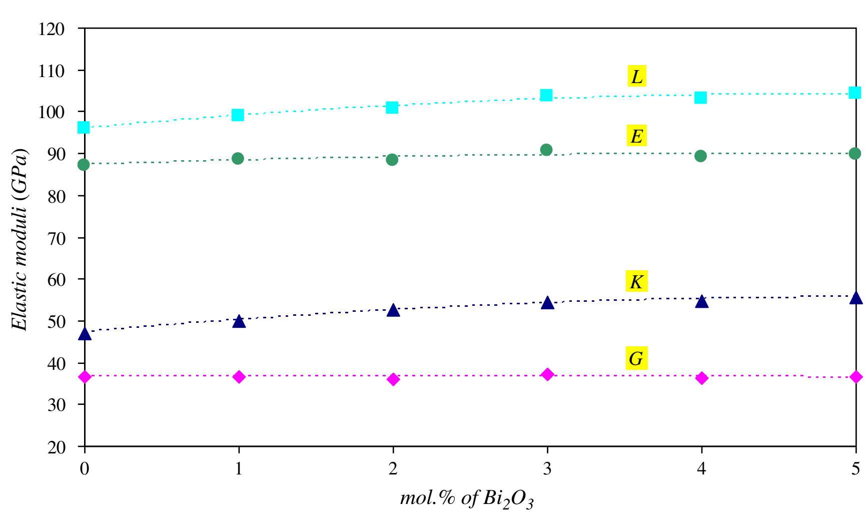

Fig. 6. Variation of elastic moduli of glass samples with different mol% of Bi2O3. The error bar is contained within the symbol size (lines are drawn as guides to the eyes).

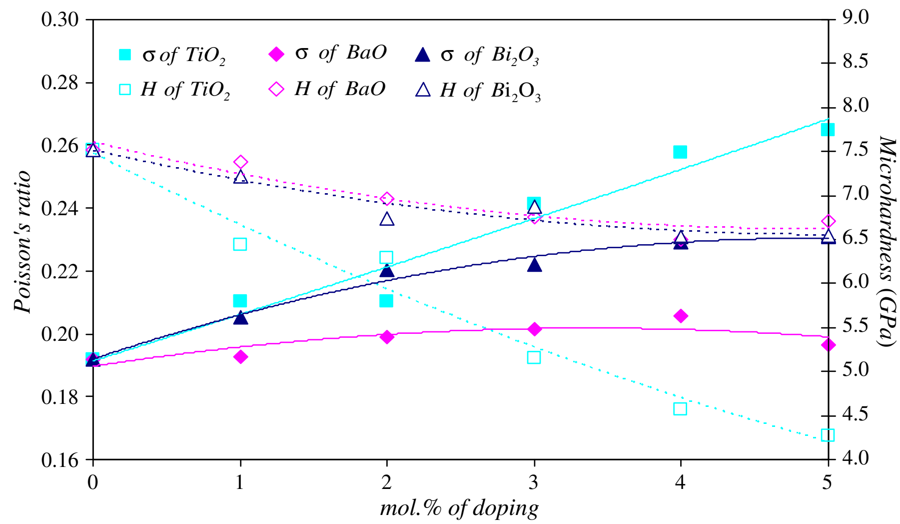

Fig. 7. Variation of Poisson's ratio (σ) and microhardness (H) of glass samples with different mol% of doping. The error bar is contained within the symbol size (lines are drawn as guides to the eyes).

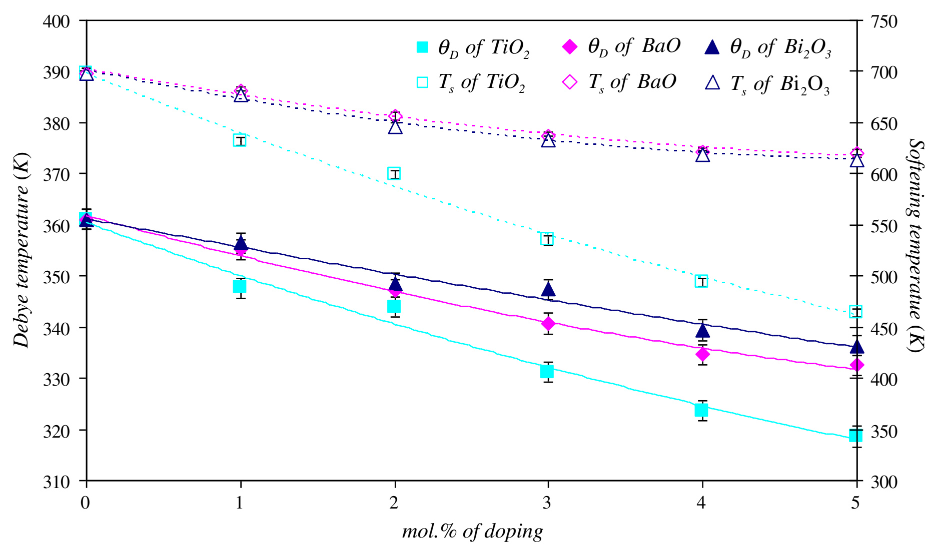

Fig. 8. Variation of Debye (θD) and softening (H) temperature of glass samples with different mol% of doping (lines are drawn as guides to the eyes).

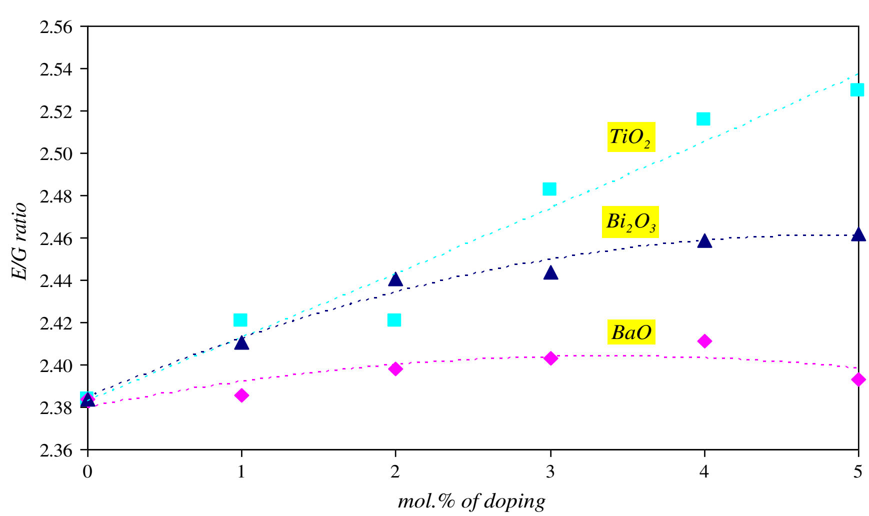

Fig. 9. Variation of E/G ratio of glass samples with different mol% of doping (lines are drawn as guides to the eyes).

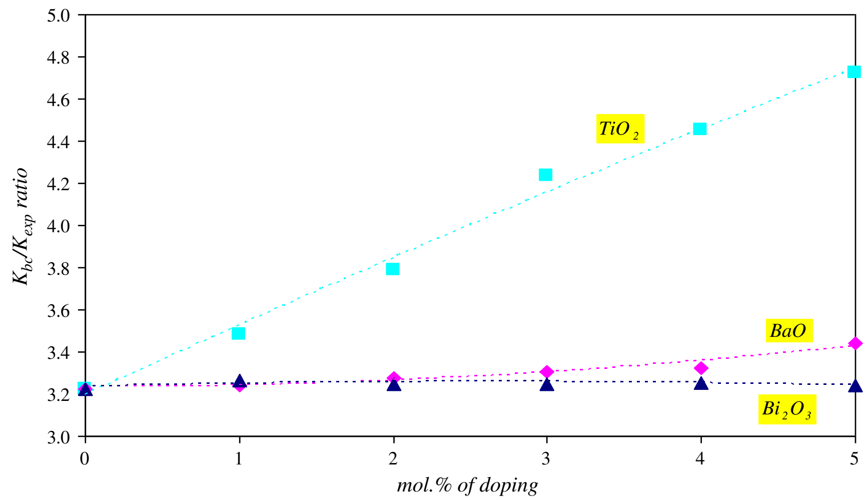

Fig. 10. Variation of Kbc/Kexp ratio of glass samples with different mol% of doping (lines are drawn as guides to the eyes).

The values of E/G, average ring diameter (l), theoretical bond compression bulk modulus (K_{bc}) and K_{bc}/K_{exp} ratio are shown in Table 3. The values of E/G were derived from the Poisson's ratio of the glasses. The relationship between E/G ratio and Poisson's ratio was related to the three chain network of the glass matrix [34]. From Table 3 and Fig. 9, it is clear that the tendency of the E/G ratio is likely the same as that of the behavior of Poisson's ratio with the increase of the mol% of the dopants. From Table 3, it is quite clear that the values of the theoretical bond compression bulk modulus (K_{bc}) decrease when the content of TiO_2 or BaO increases while K_{bc} increases with the increase of the content of Bi_2O_3. This indicates that adding HMOs to the pure composition of the glass plays a major role in the average coordination of the network structure [37] or the average stretching force constant which was found as a similar trend with the K_{bc}. In general, the ratio of K_{bc}/K_{exp} is a measure of the extent to which bond bending is governed by the configuration of the network bonds. This ratio is assumed to be directly proportional to the average ring diameter. The values of the K_{bc}/K_{exp} ratio and average ring diameter are shown in Figs. 10 and 11, respectively. From the figures, the variation of the K_{bc}/K_{exp} ratio increases as TiO_2 dopant increases while it slightly increases as the BaO dopant increases, yet it slightly decreases when the Bi_2O_3 dopant increases. Moreover, the variation of the K_{bc}/K_{exp} ratio has a similar trend to average ring diameter as shown in Fig. 11. These results of the K_{bc}/K_{exp} ratio and average ring diameter confirm the results for elastic moduli of the glass samples. Furthermore, the change in average ring diameter supports our discussion of the change of average strength of the bonds of the glass matrix with differently doped HMOs.

Table 4 shows the theoretically calculated from a theoretical bond compression model and experimental values of the elastic moduli (L, G, K and σ) for the glass samples. From the table, the calculated elastic moduli are in the range of the experimental values. It is observed that a theoretical bond compression model is in a good agreement with the experimental values of elastic moduli.

El-Mallawany et al. [32,38–41] studied the structure of TeO_2 glass which also has tetrahedral single bonds. The number of bonds per unit volume (n_b), average crosslink density ($\overline{n}_c$), ring diameter (l) and stretching force constant ($\overline{F}$) of SiO_2 are compared with those of TeO_2. These values of TeO_2 glass are listed in Table 5. The results show that the changes in the number of bonds per unit volume (n_b), average crosslink density ($\overline{n}_c$), ring diameter (l) and stretching force constant ($\overline{F}$) of TeO_2 depend on the type and the concentration of the modifiers. These results show good agreement between the changes in the structure of SiO_2 and TeO_2.

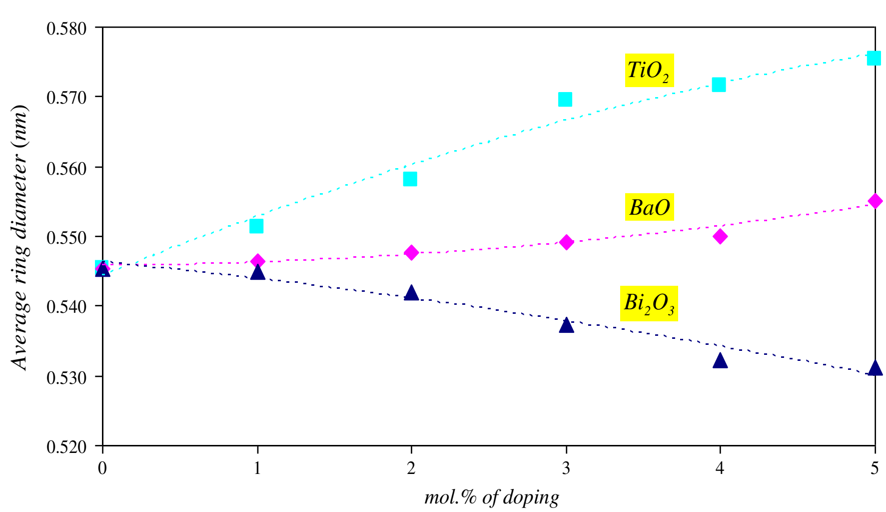

Fig. 11. Variation of average ring diameter (l) of glass samples with different mol% of doping (lines are drawn as guides to the eyes).

Table 4
Comparison of experimental estimated elastic moduli (L, G, K and σ) with those obtained theoretically by using bond compression model.

Sample No.	L_{exp} (GPa)	L_{cal} (GPa)	G_{exp} (GPa)	G_{cal} (GPa)	K_{exp} (GPa)	K_{cal} (GPa)	σ_{exp}	σ_{cal}
S0	95.87	104.02	36.56	37.37	47.12	54.32	0.1918	0.2202
S1—TiO_2	89.41	99.07	33.36	35.61	44.94	51.71	0.2025	0.2200
S2—TiO_2	86.02	93.84	32.54	33.75	42.63	48.96	0.1957	0.2197
S3—TiO_2	79.07	86.17	29.86	31.01	39.25	44.94	0.1965	0.2195
S4—TiO_2	76.20	84.45	28.29	30.40	38.47	44.02	0.2047	0.2193
S5—TiO_2	73.65	81.85	27.26	29.48	37.31	42.64	0.2062	0.2191
S1—BaO	94.73	102.84	36.06	36.91	46.65	53.75	0.1927	0.2206
S2—BaO	92.45	101.59	34.73	36.43	46.14	53.14	0.1991	0.2210
S3—BaO	90.90	100.29	33.97	35.93	45.60	52.51	0.2016	0.2214
S4—BaO	89.38	99.35	33.11	35.56	45.23	52.06	0.2057	0.2219
S5—BaO	87.82	95.54	33.17	34.16	43.59	50.11	0.1965	0.2223
S1—Bi_2O_3	97.43	103.88	37.13	37.26	47.92	54.32	0.1921	0.2209
S2—Bi_2O_3	98.73	105.58	36.96	37.81	49.45	55.29	0.2009	0.2215
S3—Bi_2O_3	101.37	108.95	37.73	38.96	51.07	57.13	0.2036	0.2222
S4—Bi_2O_3	101.31	112.52	37.47	40.18	51.35	59.08	0.2065	0.2228
S5—Bi_2O_3	102.18	113.02	37.37	40.29	52.36	59.43	0.2117	0.2235

Table 5
The number of bonds per unit volume (n_b), average crosslink density ($\overline{n}_c$),average ring diameter (l) and average stretching force constant ($\overline{F}$) of TeO_2 with different compositions [32,38–41].

Glass	$n_b \times 10^{22}$ (cm^{-3})	$\overline{n}_c$	l (nm)	$\overline{F}$ (N/m)
TeO_2	7.740	2.000	0.500	216
$80TeO_2$–$5TiO_2$–$10WO_3$–$5Nb_2O_5$	9.950	2.480	–	–
$80TeO_2$–$5TiO_2$–$12WO_3$–$3Nd_2O_3$	9.499	2.450	–	–
$80TeO_2$–$5TiO_2$–$10WO_3$–$5Nd_2O_3$	9.648	2.480	–	–
$80TeO_2$–$5TiO_2$–$10WO_3$–$5Er_2O_3$	9.684	2.480	–	–
$80TeO_2$–$20MoO_3$	8.600	–	0.512	219
$80TeO_2$–$20V_2O_5$	7.560	–	0.535	231

4. Conclusions

The elastic moduli and structure of the glass samples have been investigated as a function of compositions (different contents of TiO_2, BaO and Bi_2O_3) by measuring ultrasonic velocities. The lowest longitudinal, shear, bulk and Young's moduli of these glasses were 73.65, 27.26, 37.31 and 65.76 GPa, respectively for the composition S5—TiO_2 glass samples. The highest longitudinal, shear, bulk and Young's moduli of these glasses were 102.18, 37.73, 52.36 and 90.82 GPa, respectively for the composition S5—Bi_2O_3, S3—Bi_2O_3, S5—Bi_2O_3 and S3—Bi_2O_3 glass samples, respectively. The number of bonds per unit volume, the average stretching force constant, the average cross-link density, the average ring diameter and the theoretical bond compression bulk modulus were calculated by using a theoretical bond compression model for the asseveration of the obtained results. The agreement between the theoretically calculated and experimental elastic moduli is excellent for the studied samples. Moreover, the results showed good agreement when the number of bonds per unit volume, average crosslink density, ring diameter and stretching force constant of SiO_2 are compared with those of TeO_2. These results indicate the reliability of the experimental data.

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